

B.3 STATEMENT OF WORK (SOW)

B.3.1 Summary: This firm fixed-price requirement is for the generator and transfer switch maintenance Services at the New Jersey VA Healthcare System, specifically the Lyons VA Medical Center, 151 Knollcroft Road, Lyons, NJ 07939, and the East Orange VA Medical Center, 385 Tremont Avenue, East Orange, NJ 07018 for one (1) base year from date of award, plus provisional four (4) option years. Contractor shall provide all parts, tools, materials, labor, supervision, test equipment, technical manuals, and transportation necessary to perform **maintenance on Automatic Transfer Switches (ATS)/Manual Transfer Switches (MTS) of the Lyons campus** and **maintenance, load bank testing, and inspections of generators at both campuses.**

B.3.2 General Requirements (both campuses):

- A) Contractor work performance visits:
 - 1) Technician/s shall perform semi-annual preventative maintenance procedures on all generators.
 - 2) Technician/s shall perform annual load bank testing and preventative maintenance procedures on all generators.
 - 3) Technician/s shall perform biennial (odd numbered calendar years) preventative maintenance procedures on all generators.
- B) A field service report shall be supplied to the COR within 14 working days upon completion of each visit.
- C) All work shall comply with the latest edition of:
 - 1) Published preventative maintenance manuals from the Original Equipment Manufacturer (OEM). OEM established procedures and checklists shall be adhered to, or a contractor-supplied equivalent with approval from the Contract Officer Representative (COR).
 - 2) Joint Commission
 - 3) National Fire Protection Agency (NFPA)
 - 4) Occupational Safety and Health Administration (OSHA)
- D) Return equipment to operating conditions defined in OEM specifications.
Contractor shall ensure generators are returned to working order when work is completed (caps, valves, on/off switches, and emergency stop buttons).
- E) Contractor shall notify the COR immediately of existence or development of any defects, or repairs required, to the equipment covered under this SOW.
- F) Table 1, East Orange Campus Generator Equipment List:

Equipment Designation	Make	Catalog Number	Serial Number
Bldg. 1 A Vault 800kW	Detroit Diesel	N/A	16EE0004113
Bldg. 1 B Vault 800kW	Detroit Diesel	N/A	16E000411
Bldg. 1 C Vault 600kW	Mitsubishi	RM550	20727
Bldg. 7 650kW	Detroit Diesel	N/A	DD16VF012374

Equipment Designation	Make	Catalog Number	Serial Number
Bldg. 8 175kW	Cummins	N/A	11604063
26kV Substation CN-2	Caterpillar	SR4BGD	G4W00819
26kV Substation DN	Caterpillar	SP5	G7G0091

G) Table 2, Lyons Campus Generator Equipment List:

Equipment Designation	Make	Catalog Number	Serial Number
Gen 1G 295kW	Caterpillar	3412	3856477
Gen 2/9G 295kW	Caterpillar	3412	3856466
Gen 3/4G 700kW	Caterpillar	3412	2WJ05988
Gen 7/8G 225kW	Caterpillar	3406	9OU1111
Gen 10G 350kW	Koler/Detroit	L10123	267156
Gen 53G 230kW	Caterpillar	3406	2WB01323
Gen 54G 455kW	Caterpillar	3412	81Z22843
Gen 57G 406kW	Caterpillar	3412	81Z03544
Gen 141G 300kW	Caterpillar	C9	59P02158
Gen 143G 750kW	Caterpillar	3508	23Z06230
Gen 136IG 2,000kW	Caterpillar	3516	SBJ00750
Gen 136IIG 2,000kW	Caterpillar	3516	SBJ00751
Gen 135G 905kW	Caterpillar	C27	MJE00869
PortableG	Caterpillar	C13	NF300138
Gen 13/14G TBD	Caterpillar	TBD	TBD

- a) ***, this generator is currently in procurement process for replacement. The missing numbers shall be added later.

H) Contractors shall report to the following location for commencement of any work:

- 1) East Orange: Utility Shop, Building 1, B-Level, Room 128
- 2) Lyons: Equipment Repair Shop Foreman, Building 15, Room 6

B.3.3 Semi-Annual Generator Maintenance (both campuses):

A) Emergency Generators:

- 1) Change engine oil and oil filter.
- 2) Clean all batteries and terminals.
- 3) Check coolant system, fuel system, and lubrication systems for leaks; if necessary, add coolant and conditioner.
- 4) Change fuel filter.
- 5) Change air cleaner element.
- 6) Check turbo endplay.
- 7) Check oil level at the governor.
- 8) Clean engine crankcase breather.
- 9) Check and adjust all linkages.

- 10) Check all alarms and shutdown devices.
- 11) Clean the engine and wipe it down.
- 12) Clean all connections in the control panels and generator.
- 13) Clean all circuit boards and connectors with contact cleaner.
- 14) Drain condensation from the day tank.
- 15) Clean magnetic pickup on flywheel.
- 16) Exercise the main circuit breaker.

B) Air system:

- 1) Inspect louvers and ductwork for condition and proper operation.
- 2) Inspect air filters for condition and security; tighten clamps and brackets as required.
- 3) Inspect turbo outlet hoses and pipes for condition and security; tighten clamps and brackets as required.
- 4) Inspect and lubricate blower bypass actuators.
- 5) Inspect and lubricate the air box damper operating mechanism.
- 6) Service crankcase breathers and air box drains as required.

C) Battery sets:

- 1) Inspect battery sets for cleanliness, and clean as necessary.
- 2) Inspect terminals for tightness and cleanliness and adjust as necessary.
- 3) Check cell voltage and specific gravity of each cell and provide a report to Facility Management Service.
- 4) Check fluid levels in each cell and adjust as necessary.
- 5) Inspect battery charger and make necessary adjustments as required.

D) Controls:

- 1) Check the operation of the auto start and remote controls
- 2) Check the operation and accuracy of the AC and engine instruments.
- 3) Check operation of generator set associated indicators, lights, and alarms.
- 4) Check and adjust as required, synchronized kilowatt and reactive load sharing.
- 5) Check and adjust as required the system frequency and voltages.

B.3.4 Annual Generator Maintenance (both campuses):

A) Load bank testing

- 1) Perform load bank testing on all generators, irrespective of whether the generators reached 30% load for the year.

B) Emergency Generators:

- 1) Change engine oil and oil filter.
- 2) Clean all batteries and terminals.
- 3) Check coolant system, fuel system, and lubrication system for leaks; if necessary, add coolant and conditioner.
- 4) Change fuel filter.
- 5) Change air cleaner element.

- 6) Check turbo endplay.
- 7) Check oil level at the governor.
- 8) Clean engine crankcase breather.
- 9) Check and adjust all linkages.
- 10) Check all alarms and shutdown devices.
- 11) Clean the engine and wipe it down.
- 12) Clean all connections in the control panels and generator.
- 13) Clean all circuit boards and connectors with contact cleaner
- 14) Drain condensation from the day tank.
- 15) Clean magnetic pickup on flywheel.
- 16) Exercise the main circuit breaker.
- 17) Megger generator winding and report findings to COR.
- 18) Draw an engine oil sample for analysis and submit a written report to the Chief of Operations.
- 19) Perform load bank testing on all generators, irrespective of whether the generators reached 30% load for the year.

C) Air system:

- 1) Inspect louvers and ductwork for condition and proper operation.
- 2) Inspect air filters for condition and security; tighten clamps and brackets as required.
- 3) Inspect turbo outlet hoses and pipes for condition and security; tighten clamps and brackets as required.
- 4) Inspect and lubricate blower bypass actuators.
- 5) Inspect and lubricate the air box damper operating mechanism.
- 6) Service crankcase breathers and air box drains as required.

D) Battery sets:

- 1) Inspect battery sets for cleanliness, and clean as necessary.
- 2) Inspect terminals for tightness and cleanliness and adjust as necessary.
- 3) Check cell voltage and specific gravity of each cell and provide a report to Facility Management Service.
- 4) Check fluid levels in each cell and adjust as necessary.
- 5) Inspect battery charger and make necessary adjustments as required.

E) Controls:

- 1) Check the operation of the auto start and remote controls
- 2) Check the operation and accuracy of AC and engine instruments.
- 3) Check operation of generator set associated indicators, lights, and alarms.
- 4) Check and adjust as required, synchronized kilowatt and reactive load sharing.
- 5) Check and adjust as required the system frequency and voltages.

B.3.5 Biennial (odd numbered calendar years) Generator Maintenance (both campuses):

- A) Shall be performed in conjunction with the annual preventative maintenance procedures in section B.3.4 above, to include the following:
 - 1) Drain and flush engine cooling system with fresh water and engine coolant cleaner; change all coolant hoses.
 - 2) Replace engine thermostats
 - 3) Change engine coolant and add conditioner to new coolant
 - 4) Change all V-belts
 - 5) Change fuel line hoses on generator day tank
 - 6) Change day tank inlet fuel filter

B.3.6 Lyons Campus Only Requirements:

- A) Technician/s shall perform annual transfer switch inspections and maintenance to include ATS, MTS, and tap-boxes **only for Lyons campus**. The following buildings require work to be performed on weekends to avoid disruption of facility operations.
 - 1) Building 1: 5 ATS units
 - 2) Building 3: 4 ATS units
 - 3) Building 4: 2 ATS units
 - 4) Building 10: 2 ATS units and 1 MTS unit
 - 5) Building 135: 4 ATS units and 4 MTS units
 - 6) Building 143: 5 ATS units and 5 MTS units
- B) Annual transfer switch preventative maintenance procedures shall include, but are not limited to:
 - 1) Inspect the visually accessible portions of the transfer switch for evidence of heat, water, and mechanical damage.
 - 2) After de-energizing upstream switchgear or placing the transfer switch in bypass mode (when the building cannot be shut down), inspect the interior to verify that mechanisms are intact, together with their control circuit wiring, relays, and contacts.
 - 3) Inspect insulating parts, mechanism covers, and arc chutes for evidence of heat, discoloration, and/or mechanical damage, and replace any that are excessively worn or damaged.
 - 4) Check the alignment, deflection, gap, and wiping action of the main and arcing contacts.
 - 5) Inspect main and arcing contacts for wear, pitting, erosion, and discoloration, which indicate arcing or heat-related deterioration.
 - 6) Inspect the transfer mechanisms, coils, and contacts for evidence of damage or malfunction.
 - 7) Check the tightness of all de-energized cable, wire, and bus connections, and re-secure any loose connections that are found.

- 8) Check for the proper settings for voltage and frequency sensing, pickup, and dropout features for both normal and emergency power, as well as settings for all timing functions.
 - 9) Record and verify all settings for voltage and frequency sensing, pickup, and dropout parameters for both normal and emergency power, and settings for all timing functions. If the settings are inadvertently changed, then the recorded information will facilitate the recovery of normal operation.
 - 10) If necessary, replace eroded arcing contacts.
 - 11) Adjust main and arcing contacts for simultaneous breaking in accordance with the OEM's procedure.
 - 12) Remove old lubricant residues, then lubricate the transfer switch mechanisms and their associated devices and linkages.
 - 13) Operate the TS manually to verify proper function.
 - 14) Clean the top, bottom, and doors of the TS enclosure.
 - 15) Tap-boxes: Perform a visual inspection of the enclosure to ensure it is securely attached to its base and/or building, review conduit connections for signs of leakage, and confirm that the paint is intact with no rust or corrosion. Open all chamber doors, making sure the chambers are dry and free of debris, the gaskets are flexible with no cracking, the door hinges are secure and lubricated, and the key locks are intact and functioning. Additionally, verify that all load terminals are tightly fastened with bolts according to the manufacturer's specifications, the paint shows no signs of rust or corrosion, and the electrical connections are secure without any corrosion or cracking. Lastly, check all safety labels to ensure they are present and legible, replacing them if necessary.
 - 16) Under the coordination of the facility COR, test the re-energized ATS by switching from the normal source to the emergency source and back to verify proper functionality under load. Re-energize the MTS by switching to source two, and the facility will provide a portable generator to test the system.
- C) 2,000-hour preventative maintenance (base year, one time only)
- 1) During the first (base) year, the following generators will require 2,000 hours of planned maintenance per manufacturer standards.
 - a) Gen53G 230kW Caterpillar 3406, SN 2WB01323
 - b) Gen 57G 406kW Caterpillar 3412, SN 81Z03544

D) Table 3, Equipment List:

Equipment Designation	Make	Catalog Number	Serial Number
1ATS1	ASCO	D07ATSA30230C5XC	1950709-002 WE
1ATS2	ASCO	J00300030600C10C	862972 RE
1ATS3	ASCO	D00300030230C10C	746689-002 RE
1ATS4	ASCO	E00300030400C10C	843642-004 RE

Equipment Designation	Make	Catalog Number	Serial Number
1ATS5	ASCO	J07ATSA304005XC	2113922 WE
2ATS1	ASCO	D07ATSA30230C5XC	1950709-001 WE
2ATS2	ASCO	J0ATSA30400C5XC	1950711-002 WE
3ATS1	ZENITH	ZTS2L22EC-7	13255612
3ATS2	ZENITH	ZTS2L22EC-7	13255613
3ATS3	ZENITH	ZTSL60FC-7	13255995.1.3
3ATS4	ZENITH	N/A	115023104345
4ATS1	ASCO	J07ATSA30400C5XC	1950711-001 WE
4ATS2	ASCO	D07ATSA30230C5XC	1950709-003WE
7ATS1	ASCO	D07ATSA30230C5XC	1950709-004 WE
7ATS2	ASCO	J07ATSA30400C5XC	1950711-003 WE
8ATS1	ASCO	D07ATSA30230C5XC	1950709-005 WE
8ATS2	ASCO	J07ATSA30400C5XC	1950711-004
9ATS1	ASCO	J07ATSA30400C5XC	1950711-006 WE
9ATS2	ASCO	J07ATSA30400C5XC	1950711-005 WE
10ATS1	ASCO	J07ATBA30600C5XC	2075911WE
10ATS2	ASCO	D07ATSA30230C5XC	2056296 WE
12ATS1	ASCO	D003000301550C10C	853977-001 RE
13ATS1	ASCO	D07ATS30100C5XC	1950708 WE
14ATS1	ASCO	J0ATSA30400C5XC	1950711-007 WE
14ATS2	ASCO	J07ATSA30600C5XC	1950713 WE
14ATS3	ASCO	D07ATSA30230C5XC	1950709-006 WE
53ATS1	ASCO	J07ATS30400C5XC	1950711-008WE
53ATS2	ASCO	J07ATS30400C5XC	1950711-009 WE
54ATS1	ASCO	J07ATSA30400C5XC	1950711-010 WE
54ATS2	ASCO	D07ATSA30230C5XC	1950709-007 WE
54ATS3	ASCO	J07ATSA30400C5XC	1950711-012 WE
54ATS4	ASCO	J07ATSA30400C5XC	1950711-011 WE
57ATS1	ASCO	D07ATSA30230C5XC	1950709-008 WE
57ATS2	ASCO	D07ATSA30230C5XC	1950709-009 WE
135ATS1	ASCO	G7ACTBA31200N5X0	492136 FP
135ATS2	ASCO	J07ATBA30400N5XC	1847983 WE
135ATS3	ASCO	J07ATBA30260N5XC	1071911-002 WE
135ATS4	ASCO	J07ATBA30260N5XC	1071910 WE
136ATS1 (4,160V)	ASCO	97310445P	10445P
136ATS2 (4,160V)	ASCO	97310445P	10445P
136ATS3	ASCO	H04ATSA30800N5XC	473350-001 RE

Equipment Designation	Make	Catalog Number	Serial Number
136ATS4	ASCO	H04ATSA30800N5XC	473350-002 RE
136ATS5	ASCO	H04ATSA30800N5XC	473350-003 RE
136ATS6	ASCO	D04ATSA30030N5XC	473351-002 RE
136ATS7	ASCO	D04ATSA30030N5XC	473351-001 RE
136ATS8	ASCO	H07ATBA31000N54XC	2730807-001WE
141ATS1	ASCO	J07ATSA30400C5XC	1950712-001 WE
143ATS1	ASCO	D07ATSA30230N5XC	1950710 WE
143ATS2	ASCO	J07ATS30400N5XC	1950712-002 WE
143ATS3	ASCO	H07ATSA30800N5XC	1950714 WE
143ATS4	ASCO	J07ATSA30400N5XC	1950712-003 WE
143ATS5	ASCO	JO3MTSA3015000XC	1950717 WE
7MTS1	ASCO	JO3MTSA3023000XC	1950718-001 WE
7MTS2	ASCO	JO3MTSA3040000XC	1950720-01 WE
10MTS1	ASCO	J07AGBA30600C5XC	1950716 WE
14MTS1	ASCO	JO3MTSA3023000XC	1950720-002 WE
14MTS2	ASCO	JO3MTSA3023000XC	1950722 WE
14MTS3	ASCO	JO3MTSA3023000XC	1950718-002 WE
53MTS1	ASCO	J03MTSA3040000XC	1950720-003 WE
53MTS2	ASCO	J03MTSA3040000XC	1950720-004 WE
57MTS1	ASCO	JO3MTSA3023000XC	1950718-003 WE
57MTS2	ASCO	JO3MTSA3023000XC	1950718-004 WE
135MTS1	ASCO	J03MTSA3026000XC	1950721 WE
135MTS2	ASCO	H03MTSA3120000XC	1950724 WE
135MTS3	ASCO	JO3MTSA3040000XC	1950720-005WE
135MTS4	ASCO	JO3MTSA3040000XC	1950720-006 WE
136MTS1	ASCO	G04NTSA31600N5XC	473347-002 FP
136MTS2	ASCO	G04NTSA31600N5XC	473347-001 FP
136MTS3	ASCO	H04NTSA30800N5XC	473348-002 RE
136MTS4	ASCO	H04NTSA30800N5XC	473348-001 RE
136MTS5	ASCO	H04NTSA30800N5XC	473348-003 RE
136MTS6	ASCO	D04NTSA30030N5XC	473349-001 RE
136MTS7	ASCO	D04NTSA30030N5XC	473349-002 RE
136MTS8	ASCO	H07NTBA31000N5XC	2730806-001WE
141MTS1	ASCO	J03MTQA30400N00M	1911546 WE
143MTS1	ASCO	JO3MTSA3023000XC	1950719 WE
143MTS2	ASCO	J03MTSA3040000XC	1950720-007 WE
143MTS3	ASCO	H03MTSA3080000XC	1950723 WE

Equipment Designation	Make	Catalog Number	Serial Number
143MTS4	ASCO	J03MTSA3040000XC	1950720-008 WE
143MTS5	ASCO	J03MTSA3015000XC	1950717 WE
BLDG 7 TAP BOX	N/A	N/A	N/A
BLDG 10 TAP BOX	N/A	N/A	N/A
BLDG 14 TAP BOX	N/A	N/A	N/A
BLDG 53 TAP BOX	N/A	N/A	N/A
BLDG 57 TAP BOX	N/A	N/A	N/A
BLDG 135 TAP BOX	N/A	N/A	N/A
BLDG 141 TAP BOX	N/A	N/A	N/A
BLDG 143 TAP BOX	N/A	N/A	N/A

B.3.7 Unscheduled Repairs

- A) An allocation of five hundred (500) labor hours shall be obligated on its own line-item schedule exclusively for unscheduled repair services. These hours may be drawn upon at any time throughout the contract period of performance to address unforeseen maintenance and/or repair needs. The use of these hours shall not require prior scheduling and shall be applied as necessary to ensure timely response to emergent repair situations.
- B) An allocation of one hundred fifty thousand dollars (\$150,000.00) shall be obligated on its own line-item schedule exclusively for procurement of parts necessary to complete any unscheduled repair in accordance with this section. This allocation for parts may be drawn upon at any time throughout the contract period of performance to address unforeseen maintenance and/or repair needs. The use of these funds shall not require prior scheduling and shall be applied as necessary to ensure timely response to emergent repair situations.
 - 1) This allocation shall not be used to cover any parts, materials, or components associated with preventative maintenance activities.
- C) Emergency response time: The contractor shall respond with a phone call to the COR within two (2) hours after receipt of notification to correct the problem by phone. If the problem cannot be corrected by phone, the contractor shall commence on-site work within four (4) hours after receiving notification.